MALATHI

Student

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EDUCATION

2026
MEPCO SCHLENK ENGINEERING COLLEGE

- B.Tech Artificial Intelligence And Data Science
- 8.26 CGPA

2022
S.H.N.V HIGHER SECONDARY SCHOOL, SIVAKASI

- Higher Secondary Certificate(HSC)
- Percentage: 91.67 %

2020
S.H.N.V HIGHER SECONDARY SCHOOL, SIVAKASI

- Secondary School Certificate (SSLC)
- Percentage: 93.2%

LANGUAGES

- Tamil
- English (Fluent)
- Japanese (basic)

SKILLS

- Communication
- Creativity
- Time Management
- Team work
- Critical Thinking
- Leadership

PROFILE INFO

As a dedicated student in Artificial Intelligence and Data Science, I aim to apply my technical knowledge and creative thinking to build intelligent and scalable solutions. I am enthusiastic about solving real-world challenges through data-driven decision making and continuous learning

CERTIFICATIONS

UDEMY

Web Development ,
Mastering AWS Serverless: Hands-on With Core AWS Services

NPTEL

Python, Java, Large Language Models

INFOSYS SPRINGBOARD

Associate in IT Foundation Skills (Java, Dbms, NoSql, Software Engineering and Agile Software development)

IEEE ENGLISH FOR TECHNICAL PROFESSIONALS

AWARDS & ACHIEVEMENTS

- Awarded academic scholarship for maintaining a CGPA above 8.0
- Best Outstanding Student Award – Grade 12, for overall excellence
- Best Guide Award – Scouts (2018), recognizing leadership and service
- Scout Completion Certificate for completing full training program

PROJECTS

🧠 AI & MACHINE LEARNING & DEEP LEARNING PROJECTS

MULTI-DOCUMENT SUMMARIZATION

Python, PyTorch, PyTorch Geometric, SentenceTransformers, Neo4j

Designed a heterogeneous graph-based multi-document summarization framework integrating Graph Attention Networks trained via self-supervised BGRL, Improved Gray Wolf Optimization with Genetic Algorithm elite refinement for sentence selection, and a Graph-Enhanced PRIMERA decoder. Achieved state-of-the-art ROUGE, BLEURT, and SUPERT scores across DUC 2004, Multi-News, PubMed-MS2, and PubMed-Cochrane benchmarks.

BIOFILM FORMING BACTERIA CLASSIFICATION

Python, MobileNetV2, DCGAN, Vision Transformer (ViT), GLCM, LBP

Built an end-to-end deep learning framework to analyze, classify and predict biofilm growth in E. coli from microscopic images. Used MobileNetV2 to classify bacterial growth phases and a stabilized DCGAN to generate synthetic biofilm images to address data scarcity. Applied Otsu and adaptive thresholding for biofilm segmentation and extracted GLCM and LBP texture features to characterize biofilm structure. Designed a Time-Conditioned Vision Transformer to capture spatial and temporal biofilm growth patterns and forecast future biofilm states at unobserved time points.

STORY NARRATION GENERATOR

Python, Gradio, Cohere, Stable Diffusion, MoviePy

Created an automated storytelling platform that transforms user prompts into narrated videos using AI for text, image, and audio generation. Delivered outputs via an interactive Gradio interface.

TRUCK DELIVERY LOGISTICS ANALYZER

Python, Tkinter, Pandas, Scikit-Learn, Matplotlib

Built a desktop tool to analyze logistics data and trends with ML models like Naïve Bayes and K-Means. Provided real-time visualization of supplier performance and trip predictions. Enabled data-driven insights via a user-friendly GUI.

WEATHER PREDICTION

Python, Streamlit, Scikit-learn

Developed a web-based app for real-time weather updates and future trend prediction. Used OpenWeatherMap API with linear regression for forecasting. Integrated interactive charts and data displays using Streamlit.

MUSIC MOOD PLAYLIST GENERATOR

Python, DeepFace, Kafka, Spark, Streamlit, YouTube API

Built a real-time emotion-based playlist generator using facial recognition and Spark streaming for dynamic, privacy-aware recommendations.

🌐 WEB APPLICATIONS

JOB SEARCH PLATFORM

MERN Stack, BERT, JWT

Built a smart job portal with semantic resume-job matching using BERT and cosine similarity. Implemented role-based access, job tracking, and secure login with JWT. Streamlined hiring through automated email alerts and a modern MERN stack.

GAMIFIED HABIT TRACKER

React, Node.js, MongoDB

Designed a habit-tracking platform with gamified XP, leveling, and streak features. Enabled users to set custom challenges and earn rewards. Ensured secure authentication and responsive UX for continuous user engagement.

🎮 GAMES

SOKOBAN SOLVER GAME

Python, Tkinter

Created a puzzle game with both manual play and AI-based BFS auto-solving. Implemented real-time rendering, win logic, and animated movements. Offered multiple levels with instant reset and interactive gameplay.